Association for
Project Management

Life cycles

Learning objective:

1. Understand the distinct life cycle stages used to structure and organise a project.

APM-QS-PMQ2-3a V1

1a) Understand the distinctive features of linear, iterative and hybrid life cycles (including why projects are structured as phases in linear life cycles) and know when each is applicable.

Life cycle options and choices

There are several approaches that organisations can chose from to deliver beneficial change through project work. The choice will determine the way projects, programmes and portfolios operate and play a vital role in determining their success.

Potential approaches range from highly predictive models such as a linear approach, assuming well established knowledge in regards the context through to highly adaptive models such as iterative, which operate in more uncertain, volatile and dynamic environments.

A linear life cycle:

- Is sequential with clearly defined outputs.
- Applies to stable low-risk projects with greater structure, where expectations of each phase are known.
- Provides a clear framework for the team to follow, with early definition of requirements and allows maximum control and governance.
- Allows for controls and information to be passed on to the next phase when predefined milestones have been reached and accomplished.

Concept

Development of an initial idea, through initial studies and high-level requirements management, assessment of viability including an outline business case.

Definition

Development of a detailed definition, plans and statement of requirements that include a full justification for the work. Captured in the Project Management Plan (PMP)

Deployment

Implementation of plans and verification of performance through testing and assurance to realise intended outputs, outcomes and benefits.

Transition

Handover, commissioning and acceptance of outputs to the sponsor and wider users, culminating in formal closure.

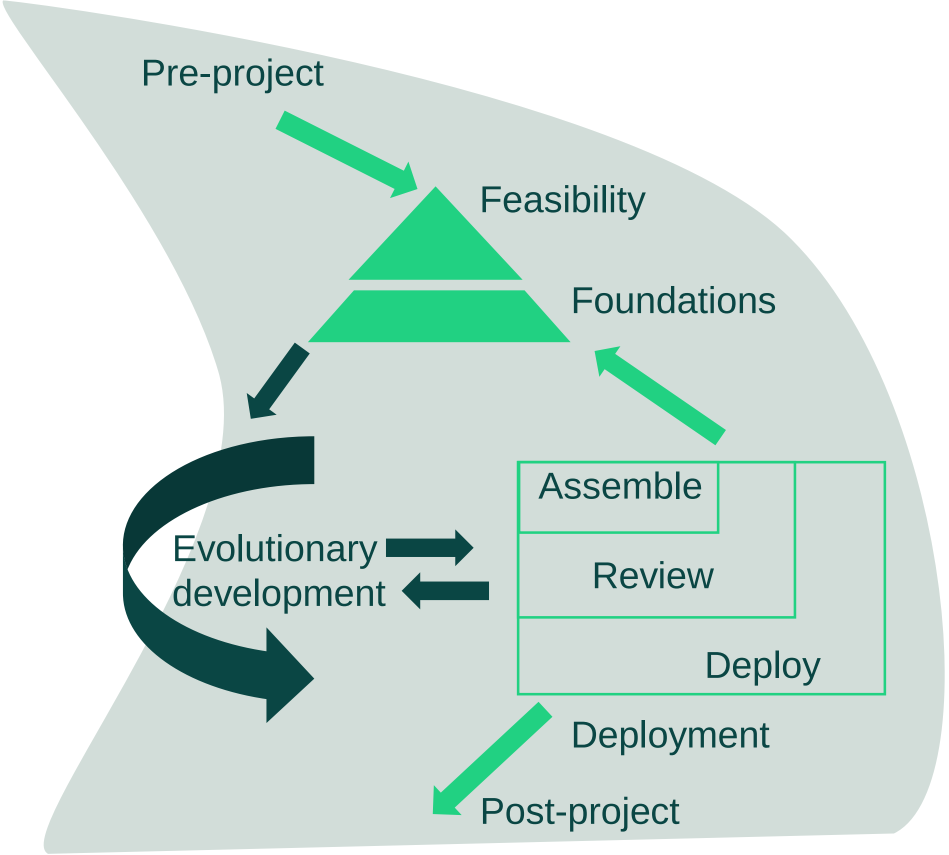
Source: Adapted from *APM Body of Knowledge*, 7th edition, Figure 1.2.2 (page 19).

Iterative life cycles

Life cycles vary according to their nature, purpose and expected use. An iterative life cycle is based on the idea of simultaneous engineering, where different development steps are allowed to be performed in parallel.

An iterative life cycle:

- Repeats one or more phases, typically multiple deployment timeboxes/sprints.
- Is beneficial for evolving objectives or solutions, used in agile development projects and allows iterative learning.
- Works if the project scope might be vague, or solutions unclear, requiring greater flexibility in governance procedures.
- Allows the project manager and team to observe the benefits that staged functionality delivers and tweak the next iteration accordingly.



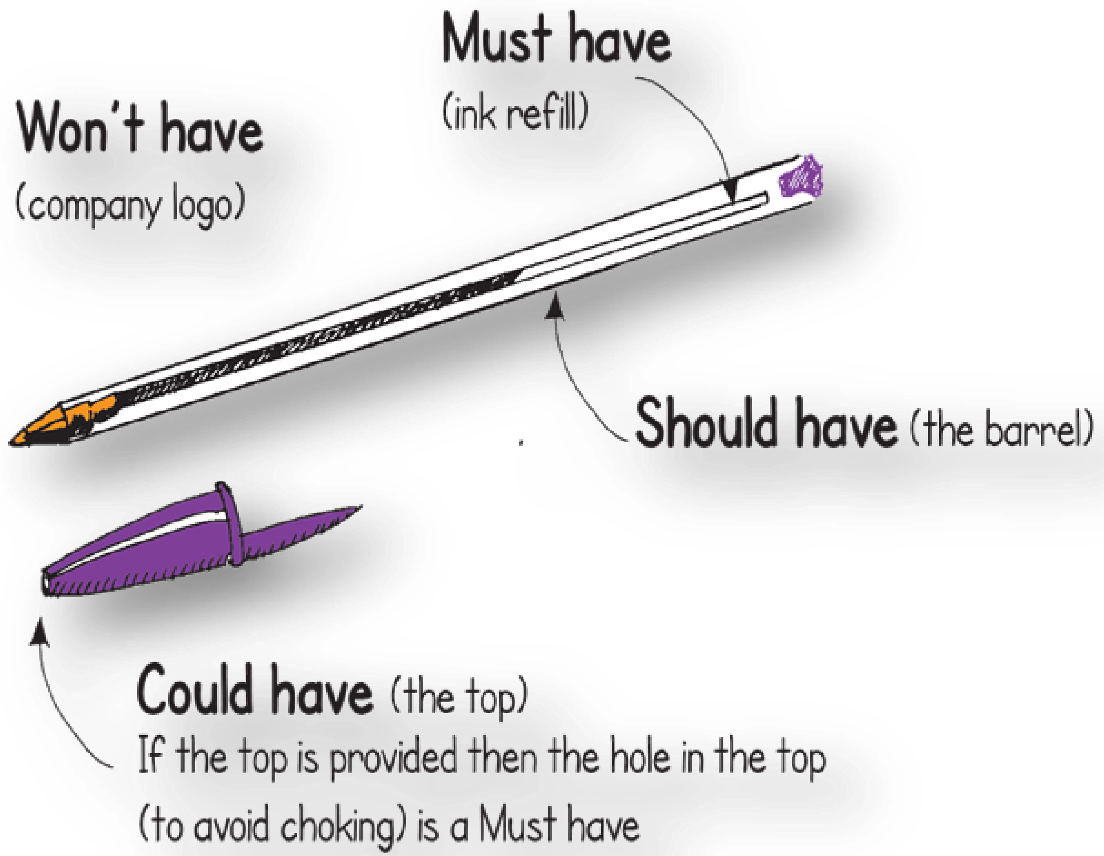
Source: Adapted from The DSDM Agile Project Framework (2014 onwards) as referenced in *APM Body of Knowledge*, 7th edition, Figure 1.2.3 (page 21)

Figure 'Life Cycles' adapted from The DSDM Agile Project Framework (2014 Onwards) https://www.agilebusiness.org/dsdm-project-framework.htmls, Agile Business Consortium. Reproduced with permission; © Agile Business Consortium, 2014.

Prioritisation techniques

Must have some flexibility
What can be sacrificed if time and cost are threatened?

MUST
O
SHOULD
COULD
O
WON'T

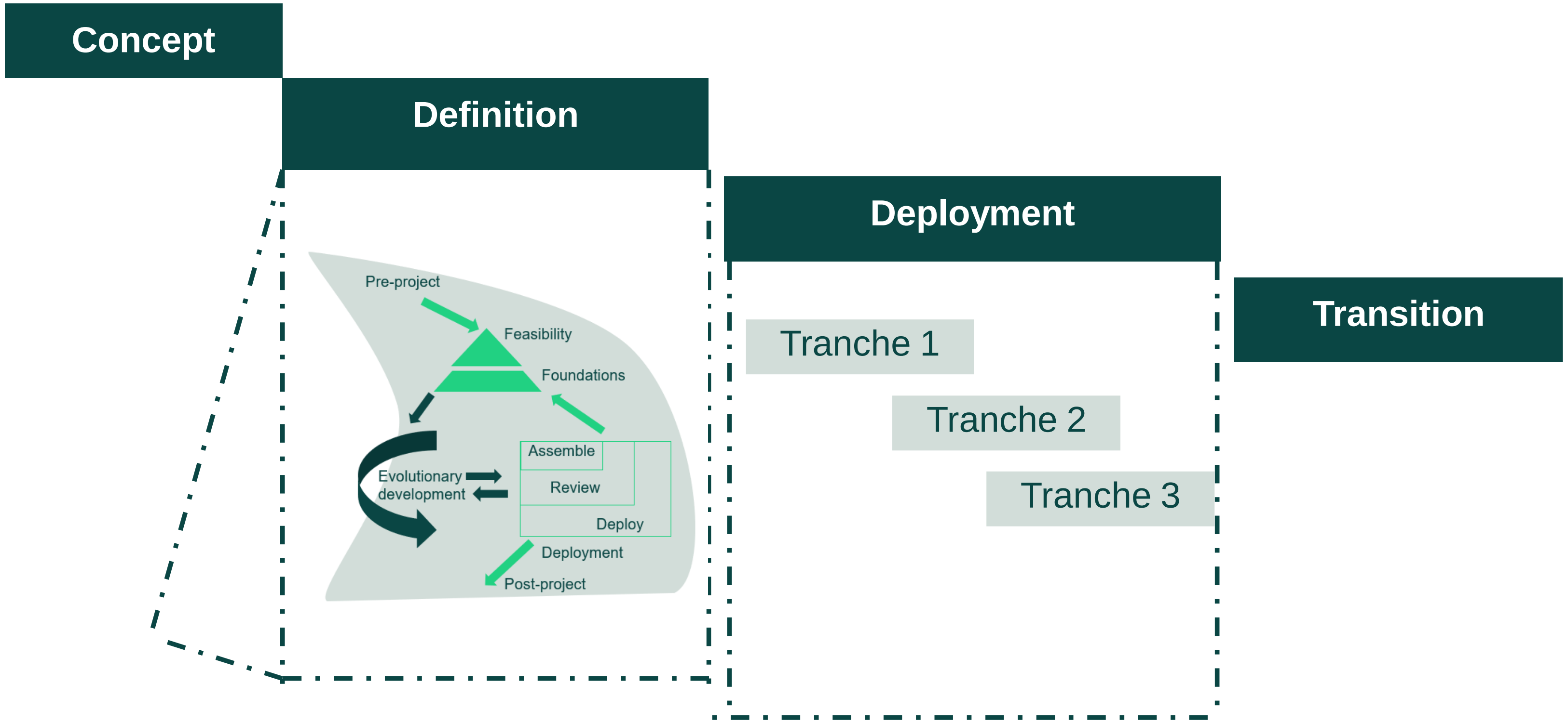


Hybrid life cycles

There is no one life cycle to suit all project contexts. Hybrid life cycles allow project managers to respond to complexities and uncertainties or the discovery of new information through iteration, by enabling a pragmatic mix of the predictive and adaptative elements of linear and iterative life cycles.

A hybrid life cycle:

- Adds iteration to a linear life cycle enabling a mix of approaches.
- Enables a project team to choose elements of both linear and iterative life cycles to best suit the project.
- Adapts – if a fixed requirement is set, the iterative style can be used during deployment to develop the output and add further functionality.



Source: Adapted from *APM Book of Knowledge*, 7th Edition, Figure 1.2.4 (page 23).
Figure 'Life Cycles' adapted from The DSDM Agile Project Framework (2014 Onwards) https://www.agilebusiness.org/dsdm-project-framework.htmls, Agile Business Consortium. Reproduced with permission; © Agile Business Consortium, 2014.

1b) Knowledge of the differences between a project life cycle and an extended life cycle.

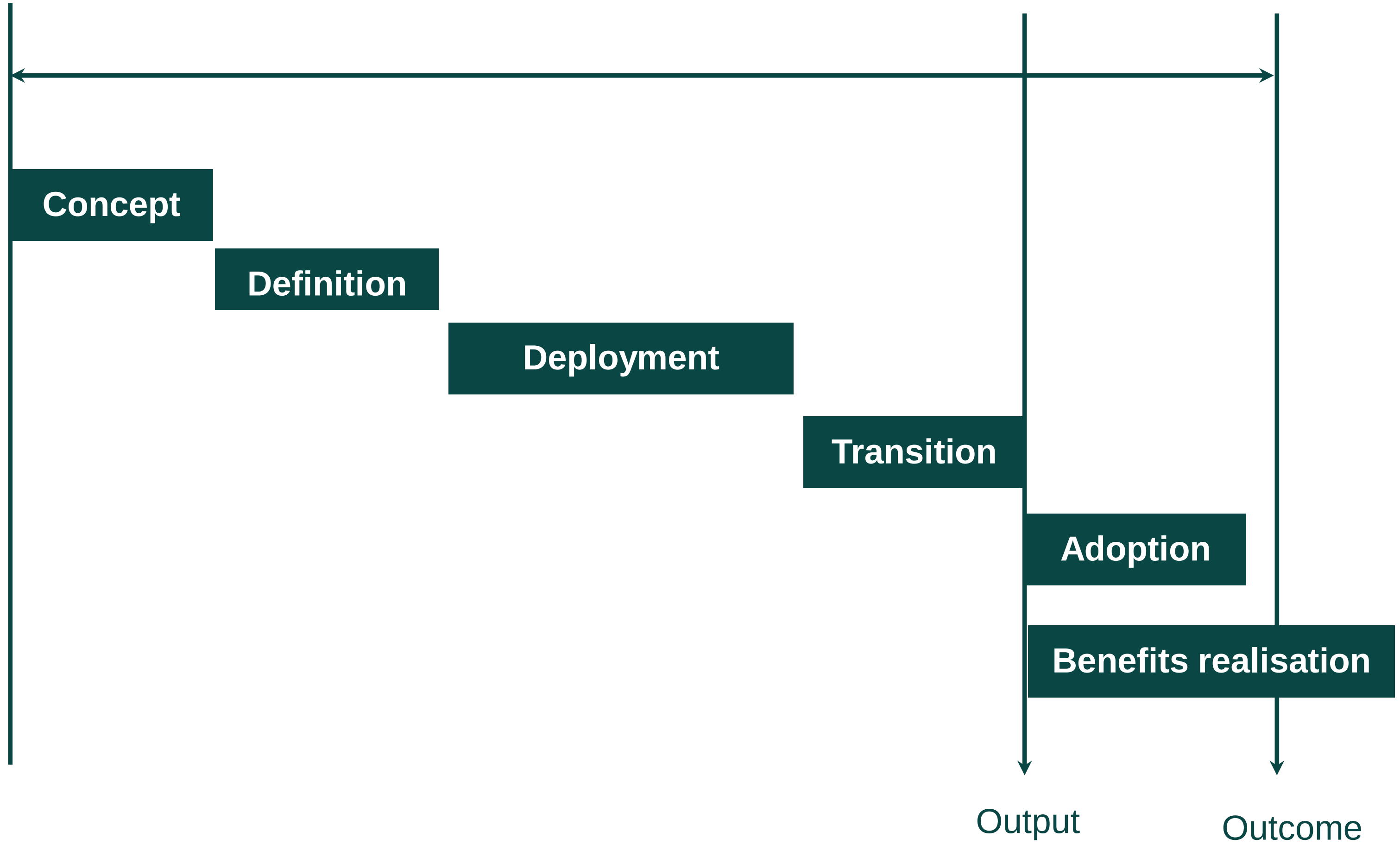
Extended life cycle

An extended life cycle goes beyond the transition and closure phase, ensuring that accountability and governance of the investment stays with the change teams until the change is fully embedded by creating a connection with benefit realisation.

An extended life cycle will therefore include the following additional phases:

- **Adoption of outputs –** operations and sustainment required to utilise the new project and enable the acceptance and use of the benefits.

- **Benefits realisation –** realisation of the required business benefits or outcomes.



Project versus extended life cycles

Before deciding on the most suitable life cycle approach, it should be decided whether a ‘project’ or an ’extended’ life cycle is suitable.

Project life cycle

- A project life cycle contains the phases up to transition where the project is closed and team disbanded.
- A business delivering the project on behalf of a customer may use the standard project life cycle, as they are likely to complete the contract at the handover of the project to the customer (end user).
- Accountability for the output is handed over, and achievement of outcomes and benefits is in the hands of the client

Extended life cycle

- An extended life cycle goes beyond the transition and closure phase to include the benefits realisation phase and adoption of outputs.
- This is suitable where the project is expected to incorporate management of change and benefit realisation.
- The extended life cycle is suited to projects that are funded within the organisation and are making a change to the business.
- The project team supports the embedding of the project.
- The team can determine the immediate benefits realised as the project goes operational and through immediate feedback to the organisation on the realisation of benefits.
- Governance and accountability for adoption of the output stays within the project until the change is fully embedded, thus preventing knowledge boundaries between project teams and operations.

Project versus extended life cycles

Aspect	Project Life Cycle	Extended Life Cycle
Documentation Requirements	Primarily focuses on documentation required for project execution and closure, such as project plans, progress reports, and final reviews.	Also requires comprehensive documentation that extends into operational manuals, long-term maintenance schedules, and end-user support materials.
Quality Assurance	Quality assurance processes are designed to ensure that the project meets defined standards and specifications at closure.	Ongoing quality assurance is crucial to maintain the integrity, performance, and safety of the project outcomes throughout their lifecycle.
Skills and Expertise	Requires skills relevant to project management and technical execution specific to the project’s requirements.	Demands a broader range of skills, including operational management, technical support, and lifecycle management expertise.
Legal and Compliance	Compliance is targeted at meeting the legal requirements up to the project's delivery.	Extends compliance to include regulatory requirements that affect the ongoing use, updates, and eventual decommissioning of the project deliverables.

1c) Understand how the context and culture of an organisation, and the needs of a specific project, influence the choice of project life cycle and any adaptations that may be needed to the life cycle.

Context, culture and project life cycles

Organisations favouring linear life cycles

- Where a highly structured, predictable and stable process is needed which offers maximum control and governance.
- Where there is availability of relatively perfect knowledge upfront.
- Which might be resistant to change and inflexible in terms of corrections and rework.
- Where knowledge / teams are divided into distinct phases, creating silos and knowledge barriers between the phases, particularly when different delivery agents will deliver different phases.
- Lower appetite for risk.

Organisations favouring iterative life cycles

- Where development projects are agile.
- Where departments are flexible, adaptable and open to change.
- They favour the idea of concurrency, or simultaneous engineering, where different development steps are performed in parallel.
- They can manage uncertainty regarding the scope by allowing the objectives to evolve throughout the life cycle as learning and discovery take place.
- Prototypes, timeboxes or parallel activities are utilised to acquire new insights, obtain feedback or explore high-risk options.

These two models alone would not suit all applications: hybrid life cycles enable a pragmatic mix of approaches to suit a specific situation.

Context, culture and project life cycles

Organisational Context and Culture (In other words, how organisational factors influence delivery of change?)

Organisational context refers to the internal and external environment in which a project operates. This includes the organisation's structure, existing processes, stakeholder expectations, and market conditions.

The following defines the influence on life cycle choice:

1. **Risk Tolerance:** Organisations with a high tolerance for risk may prefer iterative or hybrid life cycles that allow for flexibility and change. In contrast, organisations that prioritise stability and predictability might lean towards a linear life cycle.

2. **Decision-Making Style:** Hierarchical organisations may favour linear life cycles due to their clear structure and defined decision points. In contrast, organisations with a more collaborative or decentralised approach might find iterative methods more effective.

3. **Change Readiness:** The cultural readiness to embrace change influences whether an organisation can effectively implement iterative or hybrid life cycles, which require quick responses and adaptations.

Context, culture and project life cycles

Project-Specific Needs

These are the unique requirements, goals, and constraints of a project, including scope, time, cost, quality, and stakeholder expectations.

The following defines the influence on life cycle choice:

1. **Project Complexity:** More complex projects might benefit from a hybrid life cycle that combines the structured approach of linear methods with the flexibility of iterative phases.

2. **Requirement Stability:** Projects with unstable or unclear requirements are better suited to iterative life cycles, allowing for regular reassessment and realignment with project goals.

3. **Stakeholder Engagement:** Projects requiring frequent stakeholder input and feedback are likely to succeed with iterative approaches, facilitating continuous improvement and alignment with user needs.

4. **Resource Availability:** This approach allows teams to plan and execute phases where sufficient resources are assured, using the linear part of the cycle, while leveraging the iterative phases to progress with available resources.

Context, culture and project life cycles

Necessary Adaptations to Life Cycles

Purpose: Adaptations are adjustments made to the standard life cycle models to better fit the specific circumstances of the project and the organisational environment.

The following are the examples of adaptations:

1. **Phased Implementation:** In an organisation new to project management methodologies, introducing phases of a linear life cycle with checkpoints can prepare the team for more flexible approaches in the future.

2. **Custom Iterations:** For projects in dynamic sectors, customising the length and focus of iterations based on project progress and external feedback can optimise outcomes.

3. **Integration of Agile Techniques:** Incorporating agile techniques into traditional models (hybrid approach) can enhance responsiveness and stakeholder satisfaction in environments that value both innovation and control.

1d) Knowledge of the strengths and limitations of different life cycles.

Strengths and limitations of different life cycles

Linear

Strengths

- Suitable for low-risk projects.
- Provides a clear framework for the team to follow.

Limitations

- Requires very early clarity on scope and governance.
- Longer process to ascertaining benefits realised.
- Less flexible to accommodating change.

Iterative

Strengths

- Beneficial for evolving objectives or solutions, and agile projects.
- Suitable where project scope might be vague.
- Allows iterative feedback and accommodates change more easily.

Limitations

- Lack of early certainty in terms of overall duration and cost.
- Could lead to complexities in managing resource.

Hybrid

Strengths

- Facilitates a mix of approaches to suit a specific development.
- Building agile working into a project or programme can offer increased efficiency and flexibility.

Limitations

- It requires great skill and clarity when using multiple different systems of working to be successful.